



FORM TP 2020024

TEST CODE 01238010

JANUARY 2020

CARIBBEAN EXAMINATIONS COUNCIL
CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION

PHYSICS

Paper 01 – General Proficiency

1 hour 15 minutes

17 JANUARY 2020 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This test consists of 60 items. You will have 1 hour and 15 minutes to answer them.
2. In addition to this test booklet, you should have an answer sheet.
3. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer and decide which choice is best.
4. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

Sample Item

The SI unit of length is the

- (A) metre
(B) second
(C) newton
(D) kilogram

Sample Answer



The best answer to this item is “metre”, so (A) has been shaded.

5. If you want to change your answer, erase it completely before you fill in your new choice.
6. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, go on to the next one. You may return to that item later.
7. You may do any rough work in this booklet.
8. Figures are not necessarily drawn to scale.
9. You may use a silent, non-programmable calculator to answer items.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

Visualise the gravitational pull towards the earth.

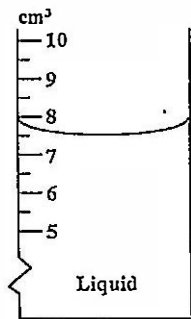
1. Which of the following instruments is suitable for measuring the diameter of a human hair?

(A) Metre rule
(B) Tape measure
(C) Vernier caliper
(D) Micrometer screw gauge

2. 0.0000462 N expressed in standard form is

(A) $4.62 \times 10^{-5} \text{ N}$
(B) $4.62 \times 10^4 \text{ N}$
(C) $4.62 \times 10^{-4} \text{ N}$
(D) $4.62 \times 10^5 \text{ N}$

Item 3 refers to the following measuring cylinder which is used to determine the volume of a liquid.



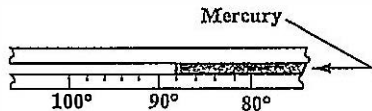
3. The volume of the liquid, in cm^3 , is

(A) 7.0
(B) 7.5
(C) 7.8
(D) 8.0

4. In an experiment to locate the centre of mass of a sheet of cardboard, a plumb line (string with a small mass on the end) is used. The plumb line is used to

(A) check that the apparatus is vertical
(B) measure the width of the cardboard
(C) show which points are vertically below the pivot
(D) check that the cardboard is swinging freely

Item 5 refers to the following diagram which shows a section of a thermometer.



5. What is the reading shown on the thermometer?

(A) 88°
(B) 89°
(C) 91°
(D) 92°

6. When standing on the moon, the mass of an astronaut is 70 kg. When he returns to earth his approximate weight will be

(A) 70 kg
(B) 420 kg
(C) 70 N
(D) 700 N

7. Which of the following is a derived unit?

(A) K
(B) s
(C) m^3
(D) kg

8. An object of mass m is attached to a spring balance and its weight, w , recorded. What will be the result if the object is taken to the moon, where the gravitational field strength is less, and weighed?

(A) Mass = m ; weight less than w
 (B) Mass less than m ; weight = w
 (C) Mass = m ; weight greater than w
 (D) Mass greater than m ; weight = w

9. A solid object, X, floats in mercury and sinks in water. A solid object, Y, floats in both mercury and water.

Which of the following statements is true about X and Y?

(A) X is less dense than Y.
 (B) X is more dense than Y.
 (C) X and Y are both denser than water.
 (D) X and Y are both denser than mercury.

10. A glass marble, X, moving with a speed of 6 m s^{-1} , collides 'head on' with an identical stationary glass marble, Y. What is the speed of Y after collision, assuming that X is brought to rest?

(A) 0 m s^{-1}
 (B) 3 m s^{-1}
 (C) 6 m s^{-1}
 (D) 12 m s^{-1}

11. Which of the following features must be present in a stable, well-designed racing car?

(A) Low centre of gravity
 (B) Narrow wheelbase
 (C) Long front
 (D) Sunroof

12. When two bodies collide, momentum is conserved. This means that the

(A) kinetic energy before impact is equal to the kinetic energy after impact
 (B) momentum of each body is unchanged
 (C) algebraic sum of the velocities before impact is equal to the sum of the velocities after impact
 (D) total momentum of the bodies before impact is equal to the total momentum of the bodies after impact

13. Which of the following expressions can be used to find the speed of an object?

(A) $\frac{\text{Change in velocity}}{\text{Time taken}}$
 (B) $\frac{\text{Change in displacement}}{\text{Time taken}}$
 (C) $\frac{\text{Distance travelled}}{\text{Time taken}}$
 (D) Distance travelled $\times$ Time taken

14. Pressure in a liquid can be calculated using the formula $P = \rho gh$.

Which of the following sets of units will give the pressure in the SI unit?

	ρ	g	h
(A)	g cm^{-3}	m s^{-2}	mm
(B)	kg m^{-3}	N kg^{-1}	m
(C)	g cm^{-3}	N kg^{-1}	m
(D)	kg m^{-3}	cm s^{-2}	cm

15. Which of the following sources of energy are classified as renewable?

I. Solar energy
II. Wind energy
III. Geothermal energy

- (A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

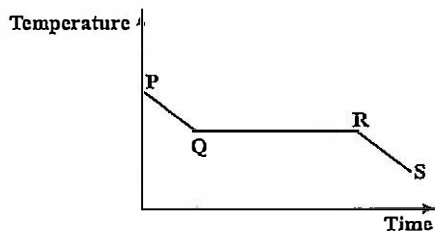
16. The heat capacity of a substance is defined as the amount of heat energy

- (A) the substance can hold
(B) 1 kg of the substance can hold
(C) required to change the substance to another state
(D) needed to raise the temperature of the substance by 1 degree

17. When a ball is thrown vertically upwards and reaches its maximum height, it has

- (A) maximum kinetic energy and maximum potential energy
(B) maximum kinetic energy and minimum potential energy
(C) maximum potential energy and minimum kinetic energy
(D) minimum kinetic energy and minimum potential energy

Item 18 refers to the following graph which is based on an experiment on change of phase.



18. The graph shows that solidification started at Q. During which of the stages is the substance in the liquid phase?

- (A) At P only
(B) Between Q and R
(C) Between R and S
(D) Between P and Q

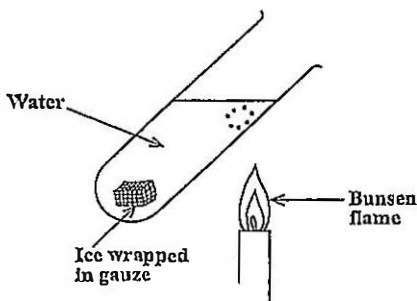
19. Which of the following processes is MOST likely to account for the fact that pot handles are usually made of wood or plastic?

- (A) Conduction
(B) Convection
(C) Evaporation
(D) Radiation

20. To check the upper fixed point on a thermometer on the Celsius scale, the thermometer is placed in
- (A) melting ice at atmospheric pressure
 - (B) boiling water at atmospheric pressure
 - (C) water in contact with melting ice at atmospheric pressure
 - (D) steam above boiling water at atmospheric pressure
21. In the pressure law, which of the following statements is true?
- (A) Pressure is constant.
 - (B) Volume is constant.
 - (C) The ratio of volume to pressure is constant.
 - (D) The ratio of volume to temperature is constant.
22. Who was responsible for arriving at the conclusion that measured amounts of electrical and mechanical energy can be converted to proportionate amounts of heat energy?
- (A) Joule
 - (B) Newton
 - (C) Rumford
 - (D) Coulomb
23. A student carries out an experiment to determine the specific heat capacity, c , of a solid using the electrical method. Which of the following equations should be used?
- (A) $c = \frac{M\Delta\theta}{PI}$
 - (B) $c = \frac{F \times D}{M\Delta\theta}$
 - (C) $c = \frac{IVt}{M\Delta\theta}$
 - (D) $c = \frac{M\Delta\theta}{IVt}$
24. Which of the following is NOT an example of evaporation?
- (A) A slice of bread left in the open air becomes dry.
 - (B) The cooling effect of sweating in animals.
 - (C) The rapid disappearance of ether if exposed to the air.
 - (D) A loaded copper wire wrapped around a block of ice gradually cuts through the ice.

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Item 25 refers to the following diagram which shows water boiling at the top of a glass test tube while a piece of ice wrapped in gauze remains unmelted at the bottom.



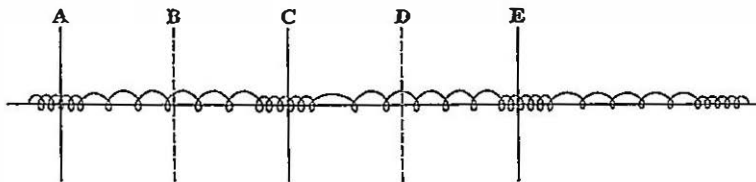
25. Which of the following statements provides the reason for this occurrence?

(A) Gauze is a poor conductor of heat.
(B) Water is a good conductor of heat.
(C) Water is a poor conductor of heat.
(D) Glass is a good conductor of heat.

26. The energy required to change the state of a substance was determined to be E_h . If the mass of the substance was DOUBLED, the new value of E_h will be

(A) doubled
(B) halved
(C) constant
(D) quadrupled

Item 27 refers to the following diagram which illustrates a longitudinal wave train produced on a slinky.



27. Which of the following distances represents the wavelength of the vibration?

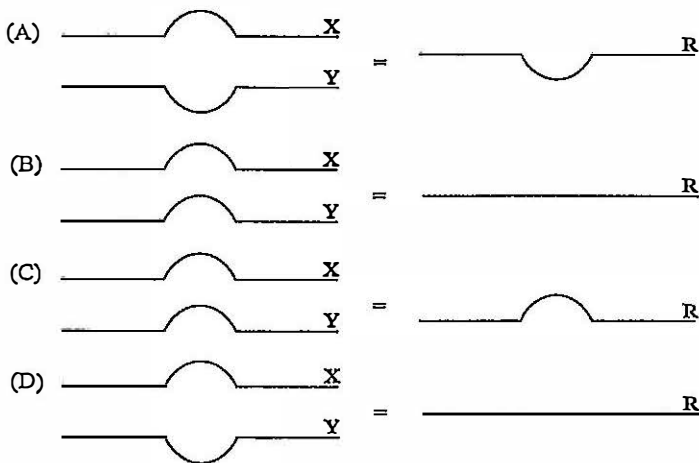
(A) A to B
(B) A to E
(C) C to E
(D) D to E

28. Which of the following electromagnetic waves has the SHORTEST wavelength?

(A) Radio
(B) Infrared
(C) Gamma
(D) Ultraviolet

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29. In the following diagrams, X and Y represent two wave pulses superposing on each other to give a resultant R. Which diagram BEST illustrates the principle of superposition?



30. If sounds of differing frequencies are played in succession, in which of the following properties would a change be detected?

(A) Pitch
(B) Speed
(C) Loudness
(D) Wavelength

31. The note from a drum is louder when it is struck harder because the sound waves produced have a greater

(A) velocity
(B) amplitude
(C) frequency
(D) wavelength

32. Which of the following are laws of refraction?

I. $\frac{\sin i}{\sin r} = \text{constant}$
 II. Angle of incidence = angle of refraction
 III. The incident ray, the refracted ray and the normal at the point of incidence all lie in the same plane

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

33. Which of the following equations, regarding units of physical quantities, is NOT correct?

- (A) $1 \text{ C} = 1 \text{ A} \times 1 \text{ s}$
 (B) $1 \text{ V} = 1 \text{ A} \times 1 \Omega$
 (C) $1 \text{ J} = 1 \text{ C} \times 1 \text{ V}$
 (D) $1 \text{ W} = 1 \text{ V} \times 1 \text{ C}$

34. A ray of light leaves air and enters glass of refractive index 1.6. The angle of refraction is 27° . What is the sine of the angle of incidence?

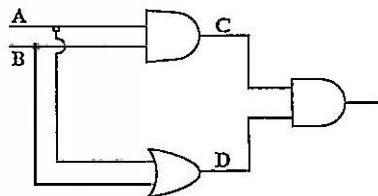
- (A) $1.6 + \sin 27^\circ$
 (B) $\frac{1.6}{\sin 27^\circ}$
 (C) $1.6 \sin 27^\circ$
 (D) $\frac{\sin 27^\circ}{1.6}$

Item 35 refers to the following table which lists the refractive indices of four different materials.

Material	Refractive Index
Air	1.0
Ice	1.3
Perspex	1.5
Diamond	2.4

35. In which material would the light waves have the SLOWEST speed?
- (A) Air
 (B) Ice
 (C) Perspex
 (D) Diamond

Item 36 refers to the following diagram.



36. What is the output at C and D when an input of 0 0 is made at A and B?

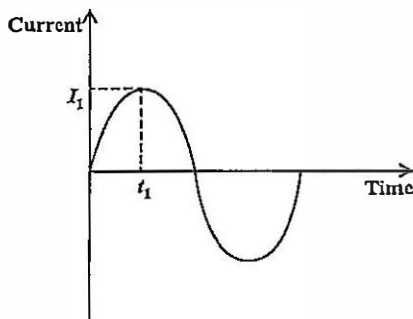
	C	D
(A)	0	0
(B)	1	1
(C)	0	1
(D)	1	0

37. Which of the following combinations of ammeter characteristics is correct?

	Resistance of an Ammeter	Connection of Ammeter to Component
(A)	Low	In series
(B)	High	In series
(C)	Low	In parallel
(D)	High	In parallel

Item 38 refers to the following graph showing the variations of alternating current with time.

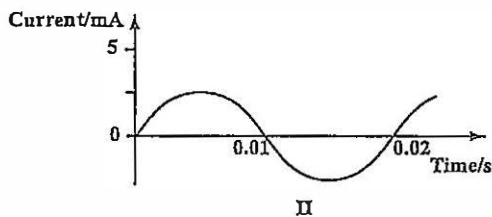
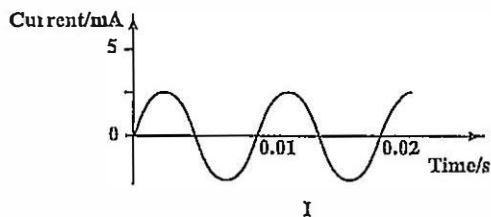
$$* V_{RMS} = \frac{1}{\sqrt{2}} \times V_{PK}$$



38. The value of the current, I_1 , at time, t_1 , is called the

- (A) root mean square value
- (B) sinusoidal value
- (C) average value
- (D) peak value

Item 39 refers to the following graphs which represent two alternating currents.



39. Which of the following combinations correctly identifies these currents?

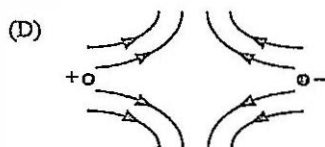
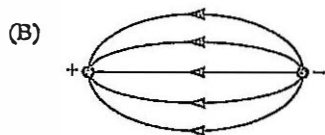
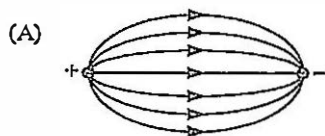
	Current with Greater Frequency	Current with Greater Peak Value
(A)	I	II
(B)	II	Neither
(C)	Neither	I
(D)	I	Neither

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40. A glass rod is rubbed with a piece of silk and becomes positively charged. The glass rod became charged by

(A) losing protons
(B) losing electrons
(C) gaining protons
(D) gaining electrons

41. Which of the following diagrams represents the electric field existing between two oppositely charged point charges?



42. A voltmeter has a very high resistance so that it can be placed in

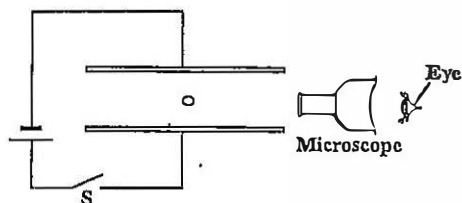
(A) parallel with a component and not affect the circuit
(B) series with a component and not affect the circuit
(C) parallel with a component and the voltmeter does not heat up
(D) series with a component and the voltmeter does not heat up

43. Which of the following statements about insulators are true?

I. All electrons are bound firmly to the atoms in an insulator.
II. In an insulator many electrons can move freely from atom to atom.
III. An insulator could be charged by rubbing.

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

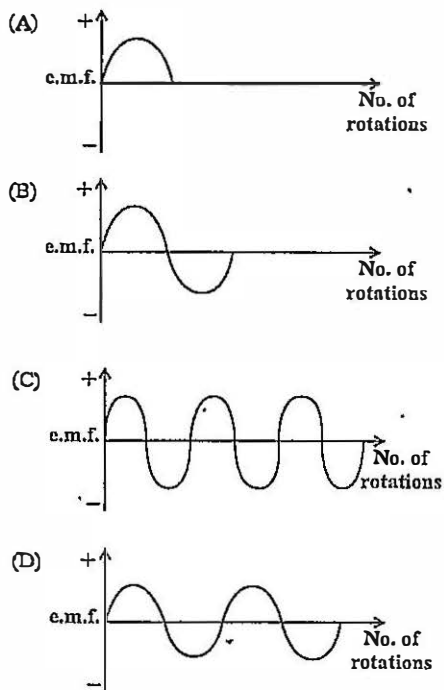
Item 44 refers to the following diagram which shows a small negatively charged dust particle midway between two plates and observed with a microscope.



44. Which of the following arrows correctly indicates the direction of the dust particle when the switch, S, is closed?

- (A) $\rightarrow$
 (B) $\leftarrow$
 (C) $\downarrow$
 (D) $\uparrow$

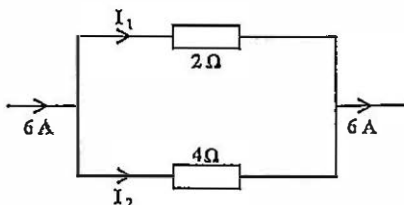
45. For a simple a.c. generator, which of the following graphs will be produced by two complete rotations of the armature?



46. Which of the following relationships is correct for the combined resistance, R_T , of resistors R_1 , R_2 , and R_3 connected in parallel?

- (A) $R_T = R_1 + R_2 + R_3$
 (B) $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$
 (C) $R_T = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$
 (D) $R_T = \frac{R_1 R_2 R_3}{R_1 + R_2 + R_3}$

Item 47 refers to the following circuit diagram.



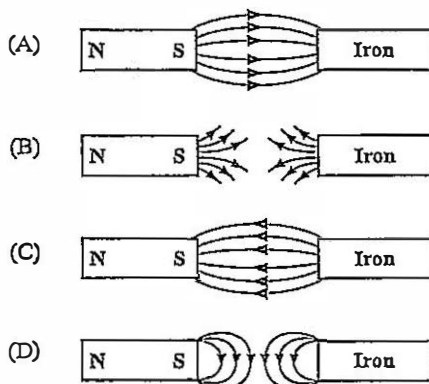
47. Using the information given in the circuit diagram, which pair of values of I_1 and I_2 is correct?

	I_1	I_2
(A)	1	5
(B)	2	4
(C)	3	3
(D)	4	2

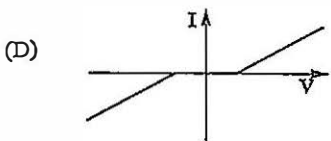
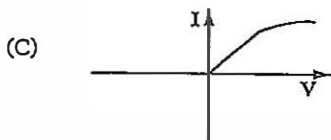
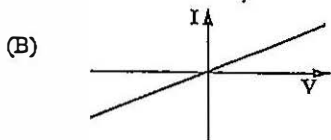
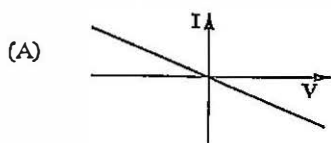
48. When an excessive current passes through a fuse, which of the following is the sequence of events?

- (A) Wire gets hot → current is cut off → wire melts
 (B) Wire gets hot → wire melts → current is cut off
 (C) Wire melts → current is cut off → wire gets hot
 (D) Wire melts → wire gets hot → current is cut off

49. Which of the following diagrams shows the magnetic field between a bar magnet and a piece of iron?



50. Which of the following diagrams represents the current/potential difference relationship for a metallic conductor at a constant temperature?



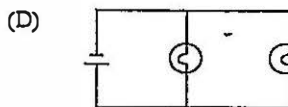
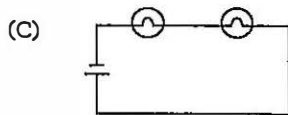
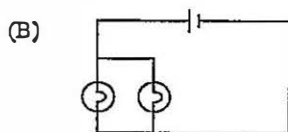
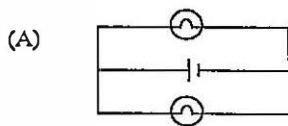
51. A device which converts sound into electricity is called

- (A) a loudspeaker
(B) a microphone
(C) an amplifier
(D) a motor

Item 52 refers to the following diagram which shows an electrical circuit with a cell and two filament bulbs.



52. Which of the following circuits are electrically the same as the circuit above?



53. Magnetic induction occurs when

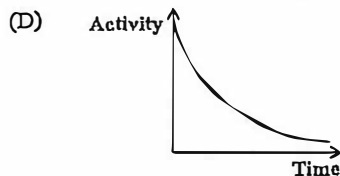
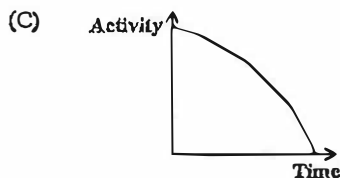
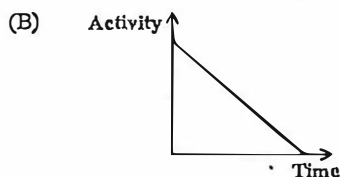
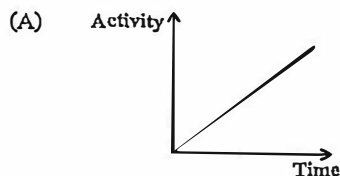
- (A) a magnet is suspended and points in the north-south direction
(B) iron nails near a magnet become magnetized
(C) a north pole attracts a south pole
(D) an electroscope is charged

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54. Isotopes of an element differ from each other in that they have different

(A) atomic numbers
(B) numbers of protons
(C) numbers of neutrons
(D) numbers of electrons

55. Which of the following graphs shows how the activity of a radioactive source varies with time?



56. According to the Rutherford-Bohr model of a neutral, stable atom, if

n = number of neutrons,
 p = number of protons, and
 e = number of electrons in the atom,

then for all elements

(A) $n = e$
(B) $p = e$
(C) $n + e = p$
(D) $n + p = e$

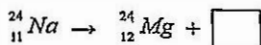
57. Which of the following symbols would be possible for an isotope of a nuclide represented by A_ZX ?

(A) ${}^{A-2}_{Z+2}X$
(B) ${}^A_{Z-2}X$
(C) ${}^{A-2}_{Z-2}X$
(D) ${}^{A+2}_ZX$

58. An isotope of uranium, ${}^{234}_{92}\text{U}$, changes to ${}^{230}_{90}\text{Th}$. This is an example of

(A) fission
(B) fusion
(C) α decay
(D) β decay

59. Sodium-24 decays into Magnesium-24 with the emission of a β -particle and can be represented by the following equation.



Which of the following particles should be placed in the box to complete the equation?

- (A) ${}^0_{-1}e$
- (B) 0_1e
- (C) ${}^4_2\text{He}$
- (D) ${}^4_{-2}\text{He}$

60. Which of the following statements about radioisotopes is true?

- (A) They are not used in scientific work because they are too dangerous.
- (B) They are most frequently used in radio communications.
- (C) They are sometimes used in medicine.
- (D) They are never found to occur naturally.

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.